VANISHING (P) PROPAGATES TO THE POSTNIKOV SYSTEM AND GUARANTEES EXTENSION OF MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12403-12457, following the statement at 12381-12401.

Editorial note. No omissions in the counted span.

Allowed standard prerequisites (editorial).

- Long exact sequence of a homological functor. lemma-representable-homological, cited at proof line 12410; complete proof at source lines 602-608, queued for separate carding. Supplies the exact Hom sequence used in the induction step of (1).
- Base case morphisms of length-one Postnikov systems. lemma-postnikov-system-small-cases, cited at proof line 12422; its span 12339-12379 partially overlaps published range 320 and its proof in the chapter is complete. Supplies the n = 1 case of (2).

Source reference. Original equation 0D80: sources/stacks-derived/pinned-original-source.tex, source lines 12265–12265. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0D81: sources/stacks-derived/pinned-original-source.tex, source lines 12340–12340. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Given a map (0D80) consider the condition*

$$\mathrm{Hom}(X_i[i-j-1], X'_j) = 0 \text{ for } i > j+1 \tag{0.1.1}$$

Then

(1) *If we have a Postnikov system for $X'_n \to X'_{n-1} \to \ldots \to X'_0$ then property (0.1.1) implies that*

$$\mathrm{Hom}(X_i[i-j-1], Y'_j) = 0 \text{ for } i > j+1$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *If we are given Postnikov systems for both complexes and we have (0.1.1), then the map extends to a (nonunique) map of Postnikov systems.*

Proof. We first prove (1) by induction on j. For the base case $j = 0$ there is nothing to prove as $Y'_0 \to X'_0$ is an isomorphism. Say the result holds for $j-1$. We consider the distinguished triangle

$$Y'_j \to X'_j \to Y'_{j-1} \to Y'_j[1]$$

The long exact sequence of Lemma 0149 gives an exact sequence

$$\mathrm{Hom}(X_i[i-j-1], Y'_{j-1}[-1]) \to \mathrm{Hom}(X_i[i-j-1], Y'_j) \to \mathrm{Hom}(X_i[i-j-1], X'_j)$$

From the induction hypothesis and (0.1.1) we conclude the outer groups are zero and we win.

Proof of (2). For $n = 1$ the existence of morphisms has been established in Lemma 0D81. For $n > 1$ by induction, we may assume given the map of Postnikov systems of length $n-1$. The problem is that we do not know whether the diagram

$$\begin{array}{ccc} X_n & \longrightarrow & Y_{n-1} \\ \downarrow & & \downarrow \\ X'_n & \longrightarrow & Y'_{n-1} \end{array}$$

is commutative. Denote $\alpha : X_n \to Y'_{n-1}$ the difference. Then we do know that the composition of α with $Y'_{n-1} \to X'_{n-1}$ is zero (because of what it means to be a map of Postnikov systems of length $n-1$). By the distinguished triangle $Y'_{n-1} \to X'_{n-1} \to Y'_{n-2} \to Y'_{n-1}[1]$, this means that α is the composition of $Y'_{n-2}[-1] \to Y'_{n-1}$ with a map $\alpha' : X_n \to Y'_{n-2}[-1]$. Then (0.1.1) guarantees α' is zero by part (1) of the lemma. Thus α is zero. To finish the proof of existence, the commutativity guarantees we can choose the dotted arrow fitting into the diagram

$$\begin{array}{ccccccc} Y_{n-1}[-1] & \longrightarrow & Y_n & \longrightarrow & X_n & \longrightarrow & Y_{n-1} \\ \downarrow & & \vdots & & \downarrow & & \downarrow \\ Y'_{n-1}[-1] & \longrightarrow & Y'_n & \longrightarrow & X'_n & \longrightarrow & Y'_{n-1} \end{array}$$

by TR3. □